

Acids, Bases and Salts — MCQ Worksheet

Science • Chapter 2 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Which indicator turns red in acid and blue in base?

- (A) Phenolphthalein
- (C) Litmus

- (B) Methyl orange
- (D) Turmeric

Q2. Phenolphthalein turns ____ in a basic solution.

- (A) Red
- (C) Pink

- (B) Yellow
- (D) Blue

Q3. All acids ionise in water to release:

- (A) OH^- ions
- (C) Cl^- ions

- (B) H^+ ions
- (D) Na^+ ions

Q4. A water-soluble base is called:

- (A) Acid
- (C) Alkali

- (B) Salt
- (D) Indicator

Q5. When dilute HCl reacts with zinc, the gas evolved is:

- (A) Oxygen
- (C) Chlorine

- (B) Hydrogen
- (D) Carbon dioxide

Q6. Acid + Metal carbonate gives:

- (A) Salt + Water + H_2
- (C) Salt + H_2 only

- (B) Salt + Water + CO_2
- (D) Salt + Cl_2

Q7. Neutralisation reaction is:

- (A) Acid + Acid $\rightarrow$ Salt
- (C) Base + Base $\rightarrow$ Salt

- (B) Acid + Base $\rightarrow$ Salt + Water
- (D) Metal + Acid $\rightarrow$ Salt

Q8. Non-metal oxides are:

- (A) Basic
- (C) Neutral

- (B) Acidic
- (D) Amphoteric

Q9. While diluting an acid, one should:

- (A) Add water to acid
- (C) Mix both equally fast

- (B) Add acid to water
- (D) Heat the acid first

Q10. The pH of a neutral solution is:

- (A) 0
- (C) 14

- (B) 7
- (D) 1

Q11. The pH of stomach (gastric) juice is approximately:

- (A) 7
- (C) 2

- (B) 9
- (D) 12

Q12. Tooth enamel begins to dissolve when pH falls below:

- (A) 7.5
- (C) 5.5

- (B) 6.5
- (D) 4.5

Q13. A bee sting can be soothed by applying:

- (A) Vinegar
- (C) Baking soda solution

- (B) Lemon juice
- (D) Milk

Q14. Acid rain has a pH:

- (A) Above 7
- (C) Equal to 7

- (B) Below 5.6
- (D) Equal to 14

Q15. The chemical name of common salt is:

- (A) Sodium hydroxide
(C) Sodium carbonate

- (B) Sodium chloride
(D) Sodium bicarbonate

Q16. Chlor-alkali process produces all EXCEPT:

- (A) NaOH
(C) H₂

- (B) Cl₂
(D) CO₂

Q17. Bleaching powder is prepared by passing chlorine over:

- (A) Dry slaked lime
(C) Limestone

- (B) Quick lime
(D) Caustic soda

Q18. Baking soda has the formula:

- (A) Na₂CO₃
(C) NaOH

- (B) NaHCO₃
(D) Na₂SO₄

Q19. Washing soda is:

- (A) Na₂CO₃
(C) NaHCO₃

- (B) Na₂CO₃·10H₂O
(D) NaOH·H₂O

Q20. The formula of plaster of Paris is:

- (A) CaSO₄·2H₂O
(C) CaCO₃

- (B) CaSO₄·½H₂O
(D) Ca(OH)₂

Q21. Plaster of Paris is obtained by heating gypsum at about:

- (A) 50 °C
(C) 200 °C

- (B) 100 °C
(D) 500 °C

Q22. The blue colour of copper sulphate crystals is due to:

- (A) Cu²⁺ ions
(C) SO₄²⁻ ions

- (B) Water of crystallisation
(D) All of these together

Q23. A salt that is acidic in nature is:

- (A) NaCl
(C) Na₂CO₃

- (B) NH₄Cl
(D) K₂SO₄

Q24. Antacids used to treat indigestion contain:

- (A) A strong acid
(C) A neutral salt

- (B) A mild base
(D) An indicator

Q25. Universal indicator gives a:

- (A) Single colour for all pH
(C) Only red colour

- (B) Range of colours for different pH values
(D) Only blue colour